

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: LAILA | Context: Arab High-School Arabic Essay Feedback (Qatar Primary)

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The input form is digital Arabic text submitted via Microsoft Forms under a 65-minute timed condition [Q25, Q26], aligning well with the deployment's text-based formative-feedback interface. The script is native Arabic (RTL), eliminating Latin-script mismatch concerns. Length distribution is unimodal around 171 words [Q134, Q135] with a smaller cluster of short essays (11–60 words, 1,061 essays) [Q136], and very few essays exceed 500 words [Q138] — a meaningful but minor mismatch with deployment, where formative drafts may vary more in length and grade-level. Model-specific input formats (AraBERT pooling + 816 features [Q156, Q157], AraT5 instruction+text [Q148], LLMs with 4096-token context [Q184]) are reasonable for Arabic NLP. Internet penetration in Qatar is 99% [WEB-14], supporting digital deployment. IF was marked LOWER priority, and the form alignment is genuinely strong; the main residual concern is that benchmark essays are exam-style timed productions while deployment drafts are pre-submission and may differ in length/polish.

ID	Evidence
Q25	"[D5] Ensuring Authentic Writing Conditions: To emulate real academic or high-stakes testing scenarios, LAILA was designed to collect essays written under timed conditions using a digital submission platform. This setup ensures consis" (p.3)
Q26	"taining one persuasive and one explanatory prompt, along with a unique, pre-generated ID to maintain students' anonymity while enabling metadata tracking [D6]. We also configured a Microsoft Form for digital essay submission, set with a 65-minute time limit to enforce a time-restricted setup [D5]." (p.5)
Q134	"The data exhibits a unimodal pattern centered around an average essay length of 171 words, with the majority of essays concentrated between 90 and 210 words." (p.15)
Q135	"A pronounced peak occurs in the 150 to 180 word range, where the count reaches its highest at 1,065 essays." (p.15)
Q136	"Shorter essays are also represented, with 1,061 essays falling between 11 and 60 words." (p.15)
Q138	"Beyond 420 words, the frequency drops sharply, with only a few essays exceeding 500 words, suggesting that longer essays are less common in the dataset." (p.15)
Q156	"The model is fine-tuned with a custom regression head and combined with handcrafted features for trait scoring." (p.16)
Q157	"The essay and prompt were provided as input to the encoder, with the max-pooled token representation concatenated with the handcrafted features." (p.16)
Q148	"The input to the model consisted of an instruction to score the essay along with the essay text, and the model was fine-tuned to generate the trait names followed by their corresponding scores." (p.16)
Q184	"The number of examples was fixed to 5, balancing scoring context with the context length limit (4096 tokens)." (p.18)
WEB-14	99% internet penetration in Qatar as of 2024 (https://www.researchgate.net/figure/Total-Number-of-Schools-in-Qatar-Primary-Intermediate-Secondary_fig1_369912994)

Strengths

- Native Arabic script, RTL-appropriate, no script-encoding mismatch [Q25, Q26]
- Digital text submission matches deployment modality
- Diverse handcrafted feature set covering surface, readability, lexical, semantic, syntactic aspects [Q54]
- Qatar internet penetration ~99% supports web-based deployment [WEB-14]

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Q54	"We implemented the 816 handcrafted features introduced by Sayed et al. (2025) for Arabic AES, encompassing surface, readability, lexical, semantic, and syntactic aspects." (p.8)
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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions match — both benchmark and deployment use digital Arabic text. Length distribution differs slightly: benchmark concentrates at 150–180 words [Q135] with few long essays [Q138], whereas deployment drafts may span a wider range.

IF-2: Yes — digital infrastructure is available; Qatar reports 99% internet penetration [WEB-14]. Secondary-geography figures are not verified [regional YAML internet_connectivity_secondary_geographies].

IF-3: Domain-specific differences: timed exam-style production [Q25] vs. formative untimed draft; benchmark lacks long-form samples (>500 words) [Q138, Q92], which limits validation for extended writing assignments.

IF-4: Minor form mismatches: length-distribution skew toward short essays and absence of long-form coverage [Q92, Q138]; timed vs. untimed production conditions.

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Q92	"Fifth, with an average essay length of only 171 words, the dataset lacks extended writing samples, restricting the assessment of models on long-form academic tasks requiring more complex, extended compositions." (p.10)

Information Gaps

- Length and polish distribution of typical formative drafts in deployment
- Internet/device infrastructure in non-Qatar target geographies

Remediation

Gap 1: Benchmark concentrates on short timed exam essays (~171 words) with very few long-form samples; deployment drafts may differ in length and polish.

Recommendation: Augment evaluation with longer-form essay samples (>500 words) and untimed draft conditions to test model robustness for formative pre-submission contexts.